

Bogdan Mocanu

Software Engineer

✉ bogdan.mocanu.gamedev@gmail.com ☎ [+40771027305](tel:+40771027305) 📍 Netherlands 🌐 [Portfolio](#) 🐙 [GitHub](#) 🔗 [LinkedIn](#)

Experience

Radboudumc

Software Developer Intern

Nijmegen

Sep. 2026 - Jan. 2027

Working on 3D reconstruction with OpenCV and Open3D libraries.

Freelance

Unity Gameplay Programmer

Remote

Jul. 2021 – Sep. 2021

Self-taught Unity (C#) gameplay programmer, freelancing on Fiverr and Upwork during summer.

Projects

Thermite | C++ Voxel Raytracing Engine

2026

- **CI/CD pipeline**: hybrid setup with Perforce and GitHub
- **Gameplay systems** (mining, weapons, barge movement)
- **Editor/Engine**: console, build packager, physics layer interface and input module
- Shipped on Steam in a **multidisciplinary team of 16** students

C++ Plugin for Godot | Grand Strategy Map Editor

2025

- Smooth map rendering with **HQX shaders** and **signed distance fields**
- **Export/import** system for grand strategy game (**Europa Universalis IV**) files
- Custom **Godot inspector** for editing country properties

Hammered | Personal C++ Simulation Engine

2025

- **Multithreaded async logging** system achieving ~11× speedup over single-threaded baseline, profiled with **Tracy**
- Dual-backend **CMake** build system generating both **OpenGL** and **Vulkan** targets from shared codebase
- **GLSL** shaders cross-compiled to **SPIR-V** at build time for both APIs

C++ Tool for Unreal Engine | Procedural Coastlines

2024

- **Wave Function Collapse** based algorithm for improving the artists' workflow
- Authored tile sets in **Tiled** (.tmx) instead of Unreal's built-in editor, based on feedback from designers
- Prototyped in **Unity**, then integrated into **Unreal Engine** as a tool used by non-programmers

Education

Breda University of Applied Sciences

Creative Media and Game Technologies

Game Engine and Tools Specialization

Bachelor of Science

Aug. 2023 - Jun. 2027

Skills

C++	Vulkan/OpenGL	Cross Platform Development	Agile Development
5 years	Using RenderDoc and Nsight	PS5, Nintendo Switch, PC	Worked in teams of 7 to 16 developers
C#	Game Engines	Engine Development	Build Systems
2 years	Unity, Godot, Unreal Engine	Creating and working with custom frameworks/engines	CMake, MSVC
Shader Programming Languages	Version Control	Concurrency	
glsl, hlsl	Git, Github, Perforce, CI/CD	multithreading, SIMD	

Languages

English

Bilingual proficiency

Romanian

Native